Pulkit **Verma**

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Research Interests

AI Safety, AI Planning, Analysis of Abstractions, and Action-Model Learning.

Education

Ph.D. in Computer Science, Arizona State University *Tempe, USA*
Advisor: Prof. Siddharth Srivastava | GPA: 4.0/4.0 *Exp. Fall 2023*

Master of Technology in Computer Science & Engineering, IIT Guwahati *Guwahati, India*
Advisor: Prof. P. K. Das | *Thesis*: Resource Usage Analysis for Speech Recognition Techniques | GPA: 9.07/10 *May 2015*

Bachelor of Engineering in Information Technology, SVITS, Indore *Indore, India*
Advisor: Prof. Anand Rajawat | *Capstone*: Smart Analyzer - Predict and Cache Webpages based on Usage. *June 2011*

Publications

Journals and Conferences

Rashmeet Kaur Nayyar*, **Pulkit Verma***, and Siddharth Srivastava. "Differential Assessment of Black-Box AI Agents". In *36th AAAI Conference on Artificial Intelligence*, 2022. (To appear). *Equal Contribution

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *35th AAAI Conference on Artificial Intelligence*, 2021.

Pulkit Verma, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016.

Pulkit Verma, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015.

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". In *International Conference on Advances In Robotics*, 2015.

Pulkit Verma, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014.

Workshops, Demos, Preprints

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *AAAI 2022 Workshop on Explainable Agency in Artificial Intelligence*, 2022. (To appear).

Naman Shah*, **Pulkit Verma***, Trevor Angle, and Siddharth Srivastava. "JEDAI: A System for Skill-Aligned Explainable Robot Planning (System Demonstration)". In submission, 2022. *Equal Contribution

Pulkit Verma. "Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems". In *IJCAI 2021 Doctoral Consortium*, 2021.

Pulkit Verma, and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI 2021 Workshop on Generalization in Planning*, 2021.

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Learning User-Interpretable Descriptions of Black-Box AI System Capabilities". In *ICAPS 2021 Workshop on Knowledge Engineering for Planning & Scheduling*, 2021.

Professional Experience

Arizona State University *Tempe, USA*
Graduate Research Associate *Aug. 2018 - (present)*

- Investigating the minimal set of requirements in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Lead a team to create an educational tool "JEDAI" to introduce AI planning concepts to laypersons using mobile manipulator robots.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

NetApp Inc.
Storage Efficiency Developer — *Bengaluru, India* — *Jul. 2015 - Jul. 2018*

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if files and volumes are cloned or modified when cloud backup is offline.

Tata Consultancy Services Ltd.
iOS Application Developer — *Pune, India* — *Mar. 2012 - Aug. 2013*

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

Teaching Experience

CSE 571 Artificial Intelligence [Graduate-level Course]
Arizona State University — *Tempe, USA* — *Fall 2019, Spring 2022*

- Delivered tutorials and readings on Robot Operating System (ROS), Constraint Satisfaction Problems (CSPs), First Order Logic, Planning Domain Definition Language (PDDL), and Dynamic Bayesian Networks.
- Co-created the software test-bed used to support the ROS-based course projects. Also created and graded homework assignments and exams.
- Mentored project groups working on course projects.

Guest Lecturer
Arizona State University — *Tempe, USA* — *2020-2021*

- CSE 574 Planning and Learning Methods in Artificial Intelligence, 2 Lectures, 39 students, Spring 2021.
- CSE 471 Introduction to Artificial Intelligence, 1 Lecture, 118 students, Fall 2020.

CS 566 Speech Processing [Graduate-level Course]
Indian Institute of Technology Guwahati — *Guwahati, India* — *Fall 2014*

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on class projects and homeworks.
- Graded homework assignments and exams.

Service

2021	**PC Member**, 32nd International Conference on Automated Planning and Scheduling (ICAPS 2022)
2021	**Reviewer**, IEEE Robotics and Automation Letters
2021	**Volunteer**, 30th International Joint Conference on Artificial Intelligence (IJCAI 2021)
2021	**Volunteer**, 31st International Conference on Automated Planning and Scheduling (ICAPS 2021)
2021	**PC Member**, ICAPS 2021 Workshop on Explainable AI Planning (XAIP'21)
2021	**PC Member**, IJCAI 2021 Workshop on Generalization in Planning (GenPlan'21)
2021	**Volunteer**, 9th International Conference on Learning Representations (ICLR 2021)
2021	**Research Grants Reviewer**, Graduate and Professional Student Association, ASU
2021	**GPSA Awards Reviewer**, Graduate and Professional Student Association, ASU
2019	**Volunteer**, Southwest Robotics Symposium 2019, Tempe, USA
2018	**Volunteer**, 16th Int. Conf. on Principles of Knowledge Representation & Reasoning (KR 2018)
2014	**Volunteer**, National Workshop on GPU Programming and Applications, IIT Guwahati

Honors & Awards

2021-2022	**GPSA Travel Grant**, Arizona State University (for AAAI 2021, AAAI 2022)
2021	**Graduate College Travel Awards**, ASU (for AAMAS 2021, ICLR 2021, ICML 2021, IJCAI 2021)
Oct 2018	**Fulton Schools Experiential Learning Grant**, Arizona State University
Aug 2013 - May 2015	**Post Graduation Fellowship**, All India Council for Technical Education, Government of India